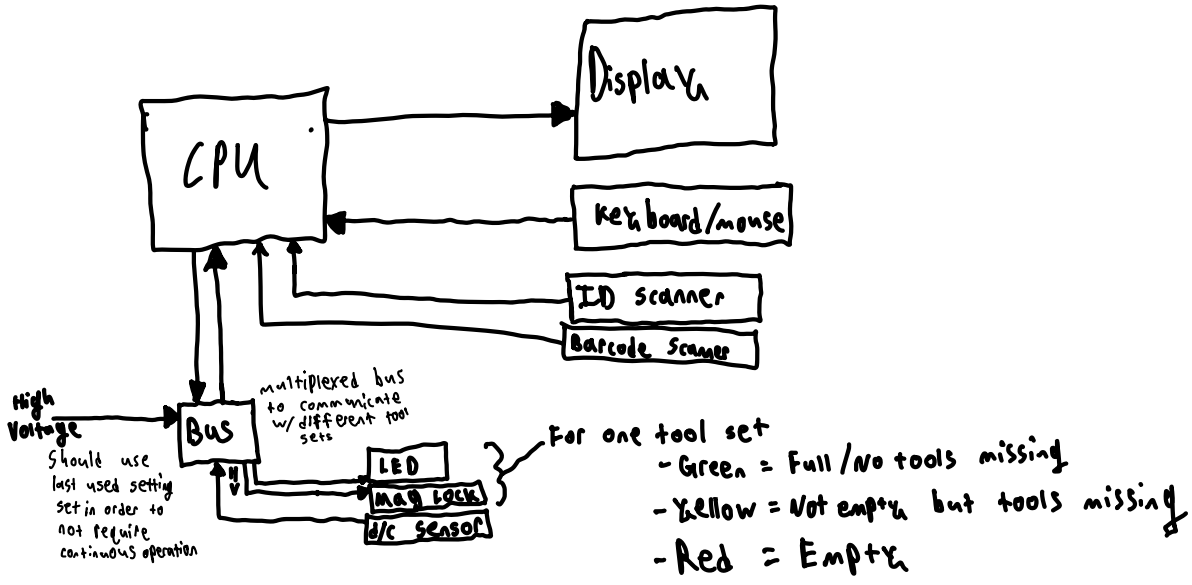
Tool organizer → uses ID to monitor tools in use & tools in storage

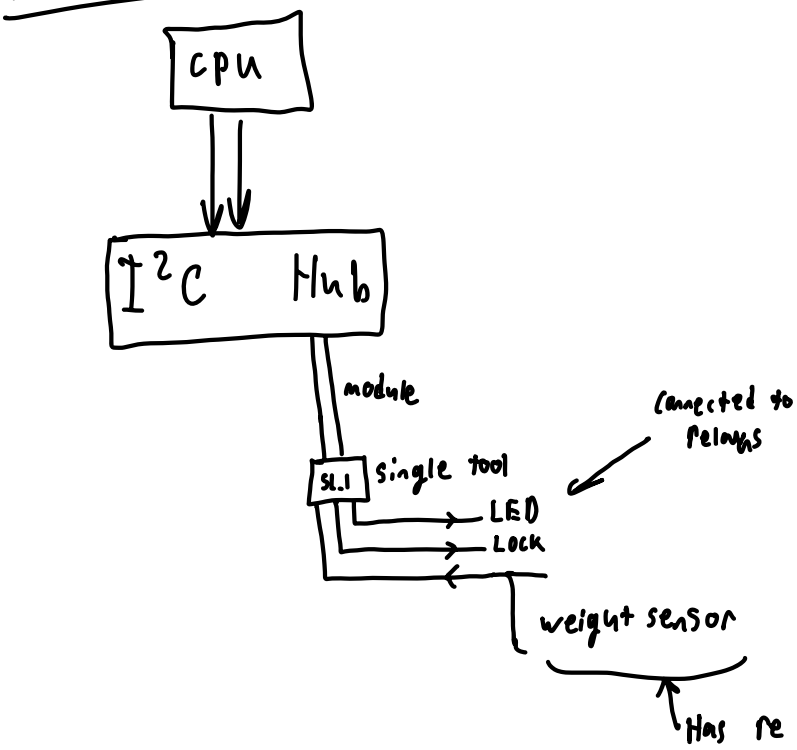
## Peripherals




**Bus** | • Set module w/ preset submit size: i.e. Bus$_{x16}$ = max of 16 tools connected

- sends size to controller to prevent overflow
- Controller can set size from 0 to Bus size
- when checking out a tool – light will flash until sensor disconnects
- Connect a sticker to each tool – scan the sticker to officially "check-out" the tool
- scan the sticker again to check it back in
- attach a weight sensor to each case to verify tool returns

module connected to the tool set




For LED to be green:

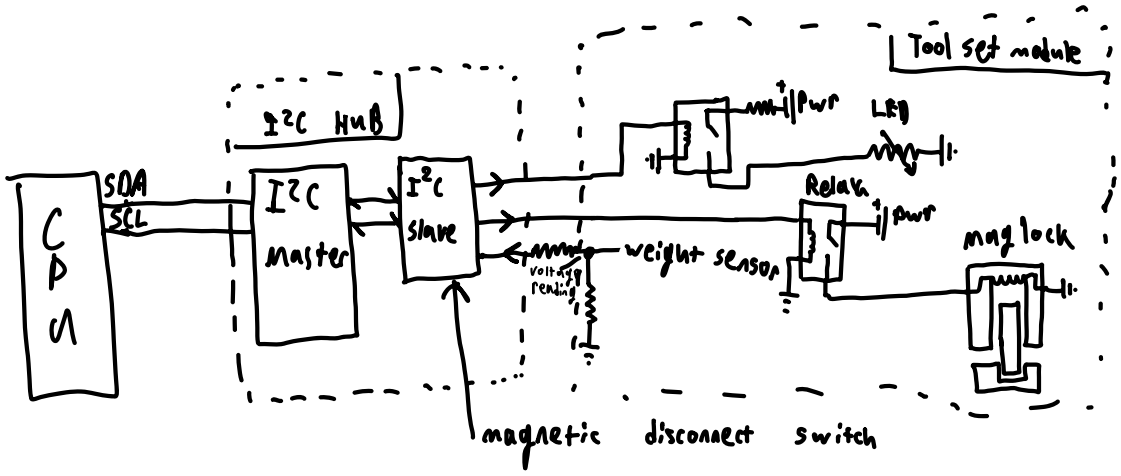
- weight sensor must be OK
- Item scan must be checked in

For LED to be yellow:

- one of the weight sensors must be gone
- one of the item scans must be checked out

For LED to be Red:

- ALL tools must be gone




I²C Hub

(PN: TXB0108

LED

PN: 564-0100-132F

LED controller

use secondary 4-channel I²C w/ a Toggle flip-flop

## Inputs

### KBM → keyboard & mouse

- for typing names of tools
- for accessing Admin settings

### ID scanner

- for checking tools in/out w/ID

### Barcode scanner

- for scanning tools in/out

# Display & UI